TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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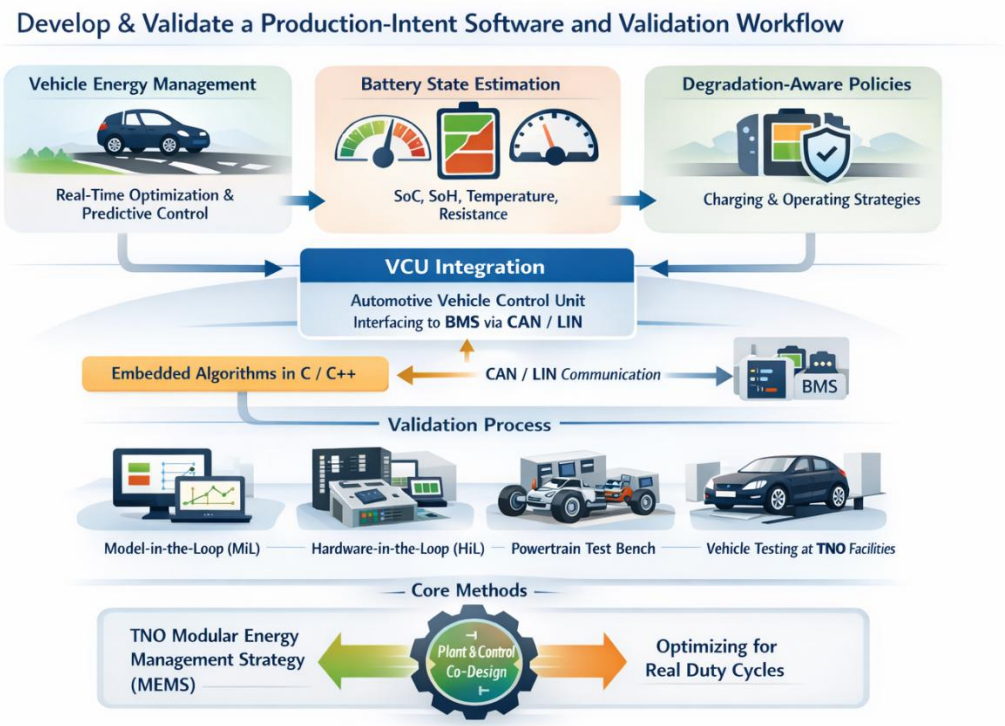
2. Technology Overview

Title of Proposed Technology	Degradation-aware Vehicle Energy Management and Battery Health Analytics for Electrification Platforms			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery & Materials		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Automotive & Railway Vehicles	Electrical and electronic devices (Code 100202)	Software	Embedded SW (Code 300801)
Concept of Proposed Technology	Develop and validate a production-intent software and validation workflow for 1) vehicle energy management (real-time optimization and predictive control), 2) battery state estimation (SoC, SoH, temperature, resistance), and 3) degradation-aware charging and operating policies. The solution targets integration into an automotive Vehicle Control Unit (VCU) and interfacing to existing Battery Management Systems (BMS) via CAN/LIN. Algorithms will be designed for embedded implementation (C/C++) and validated from model-			

	in-the-loop to hardware-in-the-loop, and finally on representative vehicles/powertrains using TNO test facilities [R5-R7].  Core methods build on TNO's Modular Energy Management Strategy (MEMS) and co-design framework to jointly optimize plant and control for real duty cycles [R3,R4,R6].
Rationale for Joint R&D Project	The Korean SME partner develops vehicle control software and electrification components and requires advanced, validated control and diagnostics capabilities to differentiate products and reduce battery-lifetime risk. TNO contributes: (a) MEMS-based control and co-design methodology demonstrated in peer-reviewed work [R3,R4,R6]; (b) degradation-aware charging and scheduling methods for EV fleets with ML and real-time optimization [R1,R2,R11]; (c) high-power battery and powertrain test infrastructure for verification at cell, module, pack and vehicle level [R5-R7]; and (d) experience with battery passport data selection and implementation considerations relevant to export-oriented products [R8,R9]. Joint R&D enables fast transfer from research prototypes to a deployable VCU/BMS software module and an evidence package (test data, models, reports) suitable for customer qualification.
Key Words	1. Modular Energy Management System (MEMS) EMS 2. SoH/SoC/temperature estimation 3. Battery degradation-aware charging 4. Embedded control (C/C++) 5. Battery passport

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	1) Implement a modular energy management controller suitable for VCU deployment, using MEMS concepts and predictive optimization [R3,R4]. 2) Develop battery state estimation modules (SoC, SoH capacity and resistance, and thermal states) compatible with BMS signals. 3) Integrate degradation-aware constraints and objectives into control and charging policies, leveraging TNO fleet-charging optimization research [R1,R2,R11]. 4) Validate end-to-end performance through MIL, SIL, HIL and physical testing (cell-to-pack and powertrain/vehicle where available) at TNO facilities [R5-R7]. 5) Deliver a documented API and reference implementation enabling the Korean SME partner to productize the functions.			
R&D Target Specifications	Embedded performance - Controller cycle time: ≤ 10 ms at 100 Hz on target ECU class (automotive-grade MCU/SoC). - Memory footprint: ≤ 10 MB RAM and ≤ 20 MB flash for full stack (configurable modules). Estimation accuracy targets (to be confirmed during requirement elicitation) - SoC: ≤ 2% RMS error over representative drive cycles (after calibration). - SoH (capacity): ≤ 5% absolute error versus reference capacity tests. - Resistance/impedance proxy: track trend with ≥ 0.9 correlation to reference pulse test metric. - Temperature (core or representative thermal state): ≤ 2°C RMS error where applicable. Degradation-aware benefits (relative to baseline SME controller) - ≥ 10% reduction in degradation-relevant stress metric (e.g., time at high SoC, high C-rate peaks, or equivalent RUL proxy) under agreed duty cycles. - Maintain or improve energy consumption (kWh/100 km or Wh/km) within ±2% while achieving the stress reduction. Validation deliverables - Test reports, calibration dataset, and traceable requirements-to-test matrix; evidence from TNO battery/powertrain test facilities [R5-R7].			
R&D Plan	**Work Package 1: Requirements and architecture (Months 1-3)** - Interfaces to SME VCU/BMS, signal list, duty cycles, constraints, safety assumptions. - Baseline controller and diagnostics benchmarking.			

	Work Package 2: Models and estimation (Months 2-10) - Battery electro-thermal and ageing model selection and calibration. - SoC/SoH/thermal estimation algorithms; validation on lab datasets. **Work Package 3: Energy management and degradation-aware optimization (Months 4-16)** - MEMS-based supervisory control design and parameterization [R3,R4]. - Add degradation-aware objectives/constraints; optional charging/fleet scheduling module [R1,R2,R11]. **Work Package 4: Implementation and integration (Months 10-22)** - Embedded C/C++ implementation; SIL and HIL testing; interface conformance. **Work Package 5: Experimental verification and demonstrator (Months 16-30)** - Cell-to-pack tests and powertrain/vehicle validation at TNO facilities; climatic and high-power scenarios as relevant [R5-R7]. - Demonstrator report and performance comparison to baseline. **Work Package 6: Dissemination, IP, and productization roadmap (Months 24-30)** - Joint IP review; documentation and training; commercialization plan led by Korean SME partner.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (research institute) expected roles and responsibilities** - System and algorithm research: MEMS-based co-design, energy management control and battery lifetime models [R3,R4]. - Charging and scheduling algorithm development, including degradation-aware optimization and ML models [R1,R2]. - Battery and powertrain testing and validation using TNO facilities (cell, module, pack and climate-controlled vehicle testing; 300 kW bi-directional cycler/simulator) [R5-R7]. - Support for battery passport-related metrics and data selection, leveraging TNO work on battery passport implementation [R8,R9]. - Knowledge transfer: joint workshops, technical documentation and support for embedded implementation.
Expected R&R of Korean SME Partner	**Korean SME partner expected roles and responsibilities** - Provide the target vehicle control unit (VCU) platform and existing vehicle control software components (e.g., drivetrain supervisory control, CAN communication, diagnostics). - Provide system requirements, representative duty cycles and operational constraints from target applications. - Implement and integrate the jointly developed control and estimation algorithms into the production-intent software stack. - Provide prototype hardware, vehicle or powertrain integration, and conduct field trials in Korea. - Lead commercialization and customer validation activities and contribute to IP generation and exploitation.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical outcomes - A validated, modular energy management and battery health estimation software stack suitable for deployment on automotive controllers. - Verified co-design workflow to select and justify powertrain designs and control parameters for specific duty cycles, based on MEMS research [R3,R4]. - Demonstrated degradation-aware charge scheduling and real-time charging strategies that improve lifetime indicators under realistic operational constraints [R1,R2]. - Laboratory-validated evidence package from cell to pack/vehicle level, including climate-controlled testing and impedance-based diagnostics where relevant [R5-R7]. Commercial and industrial impact - Differentiated controller software offering for the Korean SME partner, enabling faster electrification projects and improved battery lifetime management. - Reduced development risk through repeatable testing and model-validated design decisions. - Improved readiness for European battery passport data and metrics requirements, supporting export-oriented products [R8,R9].

5. References

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[R6] Van Putte, T. et al. "Vehicle Electrification" (2022) including battery test facility specifications (cell, module, pack, climate chambers, 750V/400A/300kW). PDF via TNO publications: https://publications.tno.nl/publication/34639492/gxMYkR/putte-2022-vehicle.pdf

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
TRL 1-2	**Basic Research Stage (1-2)**	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
TRL 3-4	**Experimental Stage (3-4)**	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
TRL 5-6	**Prototype Stage (5-6)**	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
TRL 7-8	**Productization Stage (7-8)**	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
TRL 9	**Commercialization Stage (9)**	9th Stage: Commercialization